Aliaksandr (Alex) Melnichenka

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Research interests Theoretical astrophysics and theoretical physics. Magnetized turbulence in the interstellar medium and how to measure it from radio polarization, the thermal and orbital history of icy moons, and collective instabilities of electron fluids.

Education and Training

Berea College Berea, Kentucky, **Aug 2024 to May 2027**
B.A. in Physics and Mathematics in three years, expected May 2027. GPA 3.78 (3.85 in the physics major). Dean's List every term.
Physics capstone research, full time, with Prof. Lazarian at UW-Madison Summer 2025
International Theoretical Physics Practicum (Prof. L. Levitov, MIT), admitted by graded problem sets Aug 2025
Lyceum of Belarusian State University, Minsk. Physics track of the most selective school in Belarus. Sep 2022 to Jun 2024

Publications

[1] **A. Melnichenka**, A. Lazarian, and D. Pogosyan. "Recovering 3D Magnetic Turbulence from Single-frequency Faraday Screens". *The Astrophysical Journal* 1011, 16 (2026). DOI 10.3847/1538-4357/aeac22. arXiv 2602.22204. GitHub.

[2] P. Liong*, **A. Melnichenka***, A. Bukhtatyi, A. Bilous, and L. Levitov. "Spontaneous Running Waves and Self-Oscillatory Transport in Dirac Fluids". In revision at *Physical Review Letters* (referee reports received May 2026). arXiv 2512.16571. *Equal contribution. GitHub.

[3] **A. Melnichenka**, A. Lazarian, and D. Pogosyan. "Numerical Tests of Directional Statistics and Polarization Zeros in Mixed Emitting and Faraday-Rotating Turbulence". In preparation for *The Astrophysical Journal* (2026).

[4] **A. Melnichenka**, A. Bagheri, and J. Fuller. "Has Europa Always Hosted an Ocean?" In preparation (2026).

Research Experience

Prof. Alex Lazarian and Prof. Dmitri Pogosyan Madison, Wisconsin
Undergraduate researcher and first author (Astronomy, University of Wisconsin-Madison, with U. Alberta) *Mar 2025 to present*

- First author of a paper in *The Astrophysical Journal* (published Oct 2026) that works out and tests the Lazarian and Pogosyan directional spectrum, an angle statistic that recovers the 3D magnetic turbulence spectrum from one single-frequency polarization map where Faraday tomography needs many frequencies [1].
- Built the synthetic Faraday screens and the analysis code, ran AthenaK magnetohydrodynamic cubes at 1024^3 (sonic Mach 1 and 10, over 300,000 CPU hours), drafted the manuscript and handled the referee reply with both advisors. Second paper, numerical tests of their mixed emitting and rotating theory, in preparation [3].

Prof. Jim Fuller and Dr. Amirhossein Bagheri Pasadena, California
Summer Undergraduate Research Fellow (Astronomy and Planetary Science, Caltech) *Jun 2026 to Aug 2026*

- Tested whether Europa's ocean is primordial or formed late by adding rock dehydration and an ocean-free start to Dr. Bagheri's coupled thermal and orbital model and scoring Monte Carlo histories against present-day constraints.
- Ran about 8,000 histories over the summer and tens of thousands in later parameter sweeps on a 1,000-core cluster, presented the results to Prof. Fuller's group, and now lead the tidal heating follow-up. First-author paper in preparation [4].

Prof. Leonid Levitov Cambridge, Massachusetts
Undergraduate researcher and co-first author (Physics, Massachusetts Institute of Technology) *Aug 2025 to present*

- Co-first author (equal contribution) of a manuscript predicting that a steady current in graphene breaks into self-sustained running waves, a possible narrow-band source at tens to hundreds of gigahertz. In revision at *Physical Review Letters* [2].
- Wrote the Fourier pseudospectral solver and the numerical methods section, ran the scans behind two of the figures, verified the analytic threshold and the Turing-type wavelength selection numerically, and found the regime where the wave-driven extra current turns negative. Started as my practicum team's project, one of the four winning teams.

Prof. Mark Vogelsberger and Dr. Xuejian "Jacob" Shen Cambridge, Massachusetts
Visiting undergraduate researcher (MIT Kavli Institute for Astrophysics) *Aug 2025 to Jan 2026*

- Derived velocity diffusion coefficients to third order for self-interacting dark matter near a supermassive black hole, extending Hénon's 1958 and 1960 kinetic theory, validated them by Monte Carlo over four decades in velocity, and found a catastrophic cancellation in the third order rates.
- Rebuilt the Shapiro and Marchant (1978) orbit-averaged Monte Carlo from the paper and reproduced its figures, compiled and ran a Fortran galactic nuclei code on the MIT Engaging cluster and began porting it to Python, and presented the work to the full group in December 2025.

Talks and Posters

The Magnetized Turbulent Universe: A Conference Honoring Alex Lazarian Playa del Carmen, Mexico, Nov 2025
Invited *30-minute talk on single-frequency Faraday tomography of 3D magnetic turbulence, the only undergraduate speaker*

American Astronomical Society, 247th Meeting Phoenix, Arizona, Jan 2026
Oral talk (abstract 360.07), "Single-frequency Faraday-screen tomography: a practical route to 3-D magnetic turbulence with one map"

American Physical Society, Division of Plasma Physics, 67th Annual Meeting Long Beach, California, Nov 2025
12-minute contributed talk, "Undergraduate-Led High-Resolution MHD Validation of Lazarian-Pogosyan Faraday Turbulence Theory"

Physics Congress (PhysCon), Society of Physics Students Denver, Colorado, Nov 2025
Poster, "Recovering 3-D Magnetic Turbulence from a Single Polarization Map"

Caltech SURF final presentation Pasadena, California, Aug 2026
"Has Europa Always Hosted an Ocean?"

Honors, Grants and Awards

LeRoy Apker Award nominee, Berea's candidate for the APS undergraduate research award 2026
Emergent Ventures grant, Mercatus Center at George Mason University, selected by economist Tyler Cowen 2026
Caltech Summer Undergraduate Research Fellowship award ($8,110), ten weeks with Prof. Jim Fuller 2026
Sigma Pi Sigma, national physics honor society, inducted as a sophomore 2026
Berea College conference travel awards, Career Development ($1,500, AAS 247) and Physics Department funds 2025
Putnam Mathematical Competition, rank 541 of 4,329 (top 13%), highest score in Kentucky 2025
Belarusian National Physics Olympiad, first-degree diploma (the top award) as a ninth grader (2022), second-degree (2024), third-degree (2023) 2022 to 2024
First alternate of the Belarusian International Physics Olympiad team, sixth in the national selection 2024
National Tournament of Young Physicists (Belarus), second-degree team diplomas at the finals 2023 to 2024
Phystech.International, physics olympiad run by MIPT, first-degree (2023) and third-degree (2022) diplomas 2022 to 2023
Presidential Award for Gifted Youth, Belarus 2022

Skills

Programming Python (NumPy, SciPy, Numba, h5py, Matplotlib), C++, Fortran (legacy codes), Mathematica, Git, LaTeX.
Computation Slurm clusters (MIT Engaging, U. Alberta), AthenaK MHD simulations (Kokkos, CUDA-aware MPI), pseudospectral PDE solvers, Monte Carlo methods, orbit-averaged Fokker Planck transport, polarization statistics and structure functions.
Data HDF5 and VTK simulation cubes (h5py, PyVista), FFT turbulence field generation, legacy Fortran binary readers.
Experimental Bluetooth protocol recovery for PASCO wireless sensors, photodiode and op-amp circuits, Arduino.
Languages English (fluent), Russian and Belarusian (native).

Teaching, Service and Outreach

- General Physics I and Introductory Physics I with Calculus labs, 19 and 21 students (Spring 2025). Teaching Assistant, Berea.
- Scientific Origins (general science course), astronomy labs of 15 and 17 students (Fall 2025, Spring 2026). Teaching Associate.
- Introductory Physics I and II with Calculus, grading and evening tutoring (Fall 2025, Spring 2026). Teaching Associate.
- Electricity and Electronics, Digital Electronics (Fall 2026). Lab Associate, Engineering Technology, Berea.
- C++ and algorithms, programming club, Mazyr, Belarus (2021 to 2022). Instructor, co-organized a national junior olympiad.

Savchenko Solutions savchenkosolutions.com
Founder and Lead Developer *Oct 2023 to present*

- At 16, founded the community that wrote the first complete solution manual to Savchenko's *Problems in Physics*, a 2,023-problem Russian olympiad classic. Led 102 volunteers from 23 countries, wrote or co-wrote 986 solutions. Completed Oct 2026, volunteers now make over 90% of edits. Over 100,000 readers, over 1 million page views. Featured by LEX 18 News (NBC), April 2026.

BelSO, Belarusian science olympiad platforms belso.org
Founder and Lead Developer *Sep 2023 to present*

- Built belpho.org, the first unified platform for national physics olympiad preparation in Belarus. The national jury now sends it its problems, solutions and results to publish after each olympiad. Over 80,000 visits, 75% from Belarus, every national final since 1990, and listed on the IPhO country page for Belarus. Sister sites belastro.org, belmath.org and Problempedia (2026).

Berea College Berea, Kentucky, 2024 to present
Student Government Senator at Large (elected 2026). Physics Club Astronomy Coordinator, public telescope nights (since 2025)

Selected Coursework

Blue marks Fall 2026 courses in progress.
Physics Classical Mechanics, Quantum Physics, Thermal Physics, Electromagnetic Theory, Planetary Science, Modern Physics, Advanced General Laboratory.
Mathematics Real Analysis, Topology, Combinatorics, Numerical Analysis, Differential Equations, Linear Algebra.

References

Prof. Alex Lazarian, UW-Madison, lazarian@astro.wisc.edu
Prof. Jim Fuller, Caltech, jfuller@caltech.edu
Prof. Leonid Levitov, MIT, levitov@mit.edu
Prof. Dmitri Pogosyan, University of Alberta, pogosyan@ualberta.ca